

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S6 (R,S) / S4 (PT) (R,S) Examination May 2024 (2019 Scheme)

Course Code: EET322**Course Name: RENEWABLE ENERGY SYSTEMS**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks.*

Marks

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|----|---|-----|
| 1 | Carbon monoxide is one of the most dangerous pollutants. Justify. | (3) |
| 2 | List the disadvantages of conventional energy sources. | (3) |
| 3 | Define beam radiation, diffuse radiation, and global radiation | (3) |
| 4 | Calculate the hour angle at 2:30 p.m. | (3) |
| 5 | Explain the output power versus wind speed characteristics of a wind turbine. | (3) |
| 6 | Explain the term capacity factor for a wind power plant. | (3) |
| 7 | Where do you find the occurrence of biofouling? How it can be eliminated? | (3) |
| 8 | Discuss the factors that influence the site selection for wind power plants. | (3) |
| 9 | List three types of fuel cells. | (3) |
| 10 | Write down the advantages of hydrogen energy. | (3) |

PART B*Answer one full question from each module, each carries 14 marks.***Module I**

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|----|--|-----|
| 11 | a) What do you mean by carbon credits related to the Kyoto Protocol? Which are the options to earn carbon credit? Explain the options briefly. | (8) |
| | b) Write down the disadvantages of solar power and wind power generation. | (6) |

OR

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|----|---|-----|
| 12 | a) What do you mean by the Green House Effect? List any three greenhouse gases. Explain how these gases cause Global warming. | (8) |
| | b) Write notes on the current Indian and World Energy Scenario. | (6) |

Module II

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| 13 | a) With a neat diagram explain the construction and working of a pyranometer. | (8) |
| | b) Calculate the declination angle for March 31 in a leap year. | (6) |

OR

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- 14 a) With relevant diagrams write notes on the flat plate collector and evacuated tube collector. (8)
- b) Design a solar PV system with a base load condition of 2 CFL (18 W each) and 2 fans (60 W each) working for 6 hours. Take the PV panel rating as 40 W and the operating factor as 0.75. (6)

Module III

- 15 a) With neat diagrams, differentiate between Horizontal Axis Wind Turbine and Vertical Axis Wind Turbine. Also, compare their relative advantages and disadvantages. (10)
- b) Determine the power in the wind if the wind speed is 20 m/s and the blade length is 50 m. (4)

OR

- 16 a) Discuss the various classifications of turbines for small hydro plants. (8)
- b) A wind turbine has 3 blades that are 1m long each. The air density at the site is 1.23 kg/m^3 . The wind is blowing with a velocity of 12m/s. What is the theoretical power contained in the wind? Calculate the coefficient of power of the turbine, if the maximum power output of the turbine is 392 watts. (6)

Module IV

- 17 a) A separate working fluid such as ammonia, propane, or freon is used in addition to water for closed-cycle OTEC. With a neat diagram explain the working of the closed cycle OTEC highlighting the advantage of using the additional working fluid. (8)
- b) Classify tidal power plants based on the type of basin used with relevant figures. (6)

OR

- 18 a) A hybrid cycle combines the features of both the closed-cycle and open-cycle systems. Justify. (8)
- b) List any 3 advantages and 3 disadvantages of OTEC systems (6)

Module V

- 19 a) With a neat diagram explain the construction and working of Floating Drum type or KVIC model biogas plant. (10)
- b) What are the factors that necessitate energy storage? (4)

OR

- 20 a) With a neat diagram explain the construction and working of hydrogen fuel cells. (10)
- b) Write short notes on satellite-based solar power. (4)
